

Phytotoxicity of American beech leaf leachate to sugar maple seedlings in a greenhouse experiment

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Abstract: In an effort to reexamine the factors that affect codominance of American beech (*Fagus grandifolia* Ehrh.) and sugar maple (*Acer saccharum* Marsh.), a greenhouse experiment was conducted to test the potential phytotoxic effects of American beech leaf leachate on sugar maple seedlings. We utilized an experimental protocol that addressed phytotoxic, nutrient, and pH effects by watering seedlings with seven treatment solutions: simulated throughfall water, average and dilute concentrations of beech leachate, average and dilute concentrations of maple leachate, and average and dilute concentrations of nutrient solutions. Sugar maple leachate had no effect on sugar maple seedling development, chemistry, or physiology. However, sugar maple seedlings exposed to average levels of beech leachate had significantly lower leaf area, fewer nodes, lower biomass, higher carbon assimilation rates, and higher stomatal conductance than seedlings exposed to the control solution. Results suggest that secondary compounds in beech leaf leachate are suppressing sugar maple seedling development and that field experiments are needed to explore the potential allelopathic relationship between these species.

Résumé : Dans un effort pour réexaminer les facteurs qui affectent la codominance du hêtre à grandes feuilles (*Fagus grandifolia* Ehrh.) et de l'érable à sucre (*Acer saccharum* Marsh.), nous avons réalisé une expérience en serre pour tester les effets phytotoxiques potentiels d'un lessivat de feuilles de hêtre sur des semis d'érable à sucre. Nous avons utilisé un protocole expérimental qui tient compte de la phytotoxicité, des nutriments et du pH en arrosant les semis avec sept solutions différentes : précipitation sous couvert simulée, concentrations moyenne et diluée de lessivat de hêtre, concentrations moyenne et diluée de lessivat d'érable et concentrations moyenne et diluée d'une solution nutritive. Le lessivat d'érable n'a eu aucun effet sur le développement, les caractéristiques chimiques ou la physiologie des semis d'érable à sucre. Cependant, les semis d'érable à sucre exposés à la concentration moyenne de lessivat de hêtre avaient une moins grande surface foliaire, moins de nœuds, une plus faible biomasse, un taux d'assimilation du carbone plus haut et une conductance stomatale plus élevée que les semis exposés à la solution témoin. Les résultats laissent croire que des composés secondaires présents dans le lessivat de feuilles de hêtre inhibent le développement des semis d'érable à sucre et qu'il est nécessaire de mener des expériences au champ pour explorer la relation allélopathique potentielle entre ces espèces.

[Traduit par la Rédaction]

Introduction

The question of how species maintain codominance without competitively excluding each other has been a central issue in community ecology and in forest ecology in particular (Levin 1970; Connell and Slatyer 1977; Fox 1977; Turkington et al. 1977; Ghilarov 1984). The codominance of American beech (*Fagus grandifolia* Ehrh.) and sugar maple

(*Acer saccharum* Marsh.) in the northern hardwood forest has been particularly well studied. Many researchers recognize that the relationship between sugar maple and American beech is complex and that differential responses between these two species to shade and disturbance play a major role in regeneration patterns (Fox 1977; Cypher and Boucher 1982; Kobe et al. 1995; Poulson and Platt 1996).

Although they have similar climatic ranges (Webb 1988), beech and sugar maple differ in their patterns of regeneration and response to disturbance. The uncertainty about how sugar maple and beech are able to coexist emphasizes the lack of information about how these two species interact. Assumptions about how two species maintain codominance generally focus on factors that limit the two species (Levin 1970) or on differences between adult and seedling niche requirements (Grubb 1977; Whittaker and Levin 1977). In some species, differences in regeneration patterns are based on microsite conditions, which are determined by "canopy influence" from the adult tree (Woods 1979; Woods and Whittaker 1981). Forcier (1975) suggested that small-scale disturbances provided a mechanism for coexistence and

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Table 1. Mean $\pm$ SD ion concentrations ($\mu\text{mol}\cdot\text{L}^{-1}\cdot\text{L}^{-1}$) of solutes (potassium, calcium, magnesium, ammonium, sulfate, nitrate, and chloride) in throughfall precipitation at HBEF (as reported in Lovett et al. 1996) and in the seven treatments.

Treatment	K ⁺	Ca ²⁺	Mg ²⁺	NH ₄ ⁺	SO ₄ ²⁻	NO ₃ ⁻	Cl ⁻
Throughfall	54 $\pm$ 21	18 $\pm$ 5	9 $\pm$ 2	15 $\pm$ 11	35 $\pm$ 9	27 $\pm$ 8	8 $\pm$ 4
Control	54 $\pm$ 5	18 $\pm$ 2	9 $\pm$ 2	15 $\pm$ 2	14 $\pm$ 2	27 $\pm$ 3	96 $\pm$ 10
Average beech leaf leachate	71 $\pm$ 4	24 $\pm$ 2	12 $\pm$ 2	15 $\pm$ 2	14 $\pm$ 2	35 $\pm$ 2	112 $\pm$ 11
Average maple leaf leachate	68 $\pm$ 3	26 $\pm$ 3	11 $\pm$ 2	15 $\pm$ 2	14 $\pm$ 2	33 $\pm$ 1	113 $\pm$ 14
Average nutrient	70 $\pm$ 3	25 $\pm$ 2	12 $\pm$ 2	15 $\pm$ 2	14 $\pm$ 2	34 $\pm$ 3	111 $\pm$ 12
Dilute beech leaf leachate	60 $\pm$ 4	19 $\pm$ 2	9 $\pm$ 2	15 $\pm$ 2	14 $\pm$ 2	29 $\pm$ 2	102 $\pm$ 11
Dilute maple leaf leachate	59 $\pm$ 4	19 $\pm$ 3	9 $\pm$ 2	15 $\pm$ 2	14 $\pm$ 2	28 $\pm$ 1	101 $\pm$ 14
Dilute nutrient	59 $\pm$ 4	19 $\pm$ 2	9 $\pm$ 2	15 $\pm$ 2	14 $\pm$ 2	28 $\pm$ 3	101 $\pm$ 12

Note: The ions were added as salts (CaCl₂, MgCl₂, NH₄NO₃, KCl, Ca(NO₃)₂) and then titrated to a pH of 4.6 using sulfuric acid.

demonstrated a disturbance-driven microsuccession towards beech dominance, with gaps throughout the forest in different stages of microsuccession. These two species also have different susceptibilities to disturbances, with beech being more susceptible to ice and wind damage (Melancon and Lechowicz 1987; Rhoads et al. 2002). Sugar maple's light-weight wind-dispersed seeds allow it to colonize large gaps (>400 m²) faster than beech (Hughes and Fahey 1988; Godman et al. 1990), giving this species a recolonization advantage after a disturbance. Multiple factors (e.g., nutrient availability, light levels, soil moisture, herbivory, forest disease) may be affecting the relative balance of sugar maple and beech regeneration.

There is evidence that sugar maple seedlings may be impeded by beech through both reduced light and effects of beech litter on sugar maple seedling growth (Boulaine 1962; Brisson et al. 1994; Hane 2003). This experiment investigates the phytotoxic effects of beech on sugar maple in the greenhouse as a first step in investigating a potential allelopathic effect.

Plants compete for light, water, nutrients, and (or) space, and they may also impact each other through chemical interference via allelopathy (Fisher 1980; Rice 1984). Allelopathy is the "biochemical interactions between all types of plants including microorganisms" (Rice 1984). Allelopathy is distinguished from competition, which is the removal or reduction of some resource from the environment of another plant, in that it involves the addition of a chemical to the environment of the other plant. Many ecologists use the term "interference" to encompass both of these concepts (Muller 1969; Mahall and Callaway 1992). Allelopathy has been described as a factor in the failure of tree regeneration (Gabriel 1975; Gant and Clebsch 1975; Fisher 1980), including sugar maple interacting with goldenrod (*Solidago canadensis*, *Solidago graminifolia*) and aster (*Aster novae-angliae*) in old fields (Fisher et al. 1978). Except for the extensive research addressing how black walnut (*Juglans nigra* L.) influences other species of tree seedlings, few studies have looked at allelopathic effects in cooccurring dominant tree species (Tubbs 1973; Gabriel 1975).

We designed an experiment to answer the following questions: (i) does beech leaf leachate suppress sugar maple seedling development and (ii) if so, does beech leaf leachate affect physiological processes of sugar maple seedlings that could be responsible for growth and development effects?

Materials and methods

Sugar maple seeds and beech and sugar maple leaves were collected at 500–700 m within the Hubbard Brook Experimental Forest (HBEF) in Woodstock, New Hampshire (43°56'N, 71°45'W). The vegetation of this elevation zone of HBEF is dominated by American beech, sugar maple, and yellow birch (*Betula alleghaniensis* Britt.) with some white ash (*Fraxinus americana* L.), eastern hemlock (*Tsuga canadensis* (L.) Carrière), and red spruce (*Picea rubens* Sarg.) (Bormann and Likens 1979). The forests of HBEF were selectively cut in the 1880s and 1910s and there was some salvage logging following the hurricane of 1938.

Sugar maple seeds were collected in the fall of 1998 for use in a greenhouse phytotoxicity experiment from five 2-ha areas to reduce potential environmental site-specific effects. Seeds were stored at 4°C for 4 months in plastic bags on damp filter paper (Bonner et al. 1974). Three hundred germinated seeds were planted in 4-in. pots filled with Metro Mix 350 (Scotts-Sierra Horticultural Products Co., Marysville, Ohio) in the Brown University greenhouse in February 1999.

All plants were watered every second day with a control solution that was created to match throughfall chemistry at HBEF (see treatments below and Table 1). After 2 weeks, when most plants had reached the two-leaf stage, 189 plants with two leaves between 1.0 and 4.0 cm in length were randomly assigned to one of seven treatments ($n = 189$, 27 plants per treatment). Plants were rotated on the bench weekly throughout the experiment.

The seven treatments varied in the ionic composition of the solution used to water the seedlings. The seven treatments were throughfall control and average and dilute concentrations of beech leaf leachate, sugar maple leaf leachate, and nutrients. The control treatment mimicked the composition of throughfall water at the study site (Lovett et al. 1996; mean of three hardwood stands, Table 1). By using water containing strengths of leaf leachates similar to those impacting seedlings in the field, we were able to carefully reconstruct field soil solution conditions. Many early allelopathy studies used distilled water as a control or extracting solution (e.g., Rice 1964; Tubbs 1973; Chu-Chou 1978; Fisher et al. 1978), which is likely to strip soluble ions out of the roots and soil water, which would not be the case with ion-rich soil water.

Beech and sugar maple leaf litter soaked in the control solution and then diluted with control solution to in situ dissolved organic carbon (DOC) concentrations was used to test phytotoxic effects on sugar maple seedlings. The relative amount of soil water DOC at the bottom of the Oa horizon that is derived from leaf leachate is not well known, but it has been estimated to be <30%, so we tested both 30% (average treatments) and 5% (dilute treatments) (McDowell and Fisher 1976; McDowell and Likens 1988; Johnson et al. 1995). The beech treatments (average beech, dilute beech) were used to test for effects of beech secondary compounds on sugar maple seedling development. The maple leachate treatments on sugar maple seedlings were used as a procedural control to separate out generic DOC effects from specific species-related DOC effects. To isolate the effects of DOC from nutrient additions and pH effects, we included two inorganic nutrient treatments (average and dilute nutrient concentrations) with the same nutrient concentrations as the leaf leachate solutions but none of the secondary compounds. To separate nutrient and species effects, we did planned contrasts of three treatments in the following combinations: average beech, average nutrient, control; dilute beech, dilute nutrient, control; average maple, average nutrient, control; and dilute maple, dilute nutrient, control.

The leachate solutions were made using sugar maple and beech leaves from the study areas collected from under the snow in February 1999. The leaves were transported to the laboratory packed in snow and then sorted by species and soaked in simulated throughfall solution at 4°C for 2 days. The DOC concentration of the resulting leachate was determined after filtering (0.45-mm filter) by persulfate oxidation of organic matter followed by infrared CO₂ detection (Dohrmann Instruments, Xertex Co., Santa Clara, Calif.). The leachate was diluted to 30 and 5% of the average Oa soil water DOC concentration reported for HBEF (McDowell and Fisher 1976; McDowell and Likens 1988; Johnson et al. 1995). All leachates were stored in a cold room at 4°C for the duration of the experiment.

The two nutrient treatments were created by measuring the nutrient concentrations (ion chromatograph; Dionex Corporation 1999) in the average and dilute leachate solutions of beech and sugar maple. Concentrations of potassium, sodium, magnesium, calcium, sulfate, and nitrate were replicated in the solution (Table 1). Chloride was used to compensate for any anion imbalances, as it is not a limiting plant nutrient. Although the concentrations of chloride were elevated well above in situ levels, the concentrations were still well below toxic levels (Epstein 1972). All solutions were titrated to a pH of 4.6 with sulfuric acid to match hydrogen concentrations of throughfall water at the study site (Table 1) (Lovett et al. 1996). Since sulfate salts are generally highly insoluble, and we did not want the salts being deposited into the soil in undissolved form, all of the sulfate was added as sulfuric acid, which limited the amount that could be added without depressing the pH below 4.6. However, even at 14 µmol·L⁻¹, sulfate is unlikely to be a limiting nutrient for plant growth relative to the other nutrients (Epstein 1972).

Every second day, each seedling was watered with the appropriate solution until water ran out of the bottom of the

pot. The plant height, leaf length, and number of leaves were recorded weekly for each plant. During the 10th week of the experiment, carbon assimilation rates and stomatal conductance were measured for the seedlings in the average beech and average nutrient treatments. These measurements were made on the first and second set of leaves of each seedling using an infrared gas analyzer (ACD LCA4, environmentally controlled). At the end of 12 weeks (May 18), the plants were harvested, leaf area was measured using a LI-COR LI-3100 area meter, and a random subsample ($n = 17$) of seedlings from each treatment was dried at 50°C for biomass measurements.

The leaves from 10 plants and stems and roots from three plants were randomly selected from each treatment for nutrient analysis. The plant material was dried at 50°C and then ground. Nitrogen content was determined using the modified Dumas technique (Bremner 1996). Calcium, magnesium, and potassium were determined by digesting 0.1 g of sample in 70% nitric acid by microwaving for 10 min at 210°C (Milestone Ethos 1600 microwave digestion unit), diluting to a 10% nitric solution, and then determining concentrations using ICP-ES (Perkin Elmer Optima 3000). Root digests were filtered (0.45-mm filter) to remove any undissolved soil particles. Peach (NBS No. 1547) and pine (NBS No. 1575) leaf reference material was used to estimate accuracy and precision for both the CN analyzer and the ICP. The peach and pine reference materials for nitrogen, magnesium, and calcium were all within 1–2% of certified values. Potassium had more variability and was within 5%. Repeated samples of the same material were all within 2% of each other.

Statistical methods

One seedling died during the greenhouse experiment, and it was excluded from all analyses. Seedling growth rates, biomass, leaf area, and tissue chemistry were analyzed using planned contrasts of groups of three treatments. Treatment was used as a fixed factor in a one-way ANOVA for each group of three treatments. All variables were checked for normality using the Shapiro-Wilk W test and log transformed if needed. Tukey–Kramer post hoc comparisons among the groups of three treatments were carried out to test for differences among the three treatments within each planned contrast group. Carbon assimilation rate and stomatal conductance were tested for time of day effects, and no significant time of day effect was found (carbon assimilation rate: $p = 0.34$; stomatal conductance: $p = 0.47$), so no adjustments for time of day were necessary (Farris and Lechowicz 1990). Due to colinearity of the variables of leaf area and number of nodes, leaf area alone was used as a predictor of the rate of carbon assimilation (A), stomatal conductance (g), and water use efficiency (A/g). Treatment was also used in an ANOVA to predict carbon assimilation, stomatal conductance, and water use efficiency. Treatment and leaf area were not used together in a single model because of the colinearity between these variables. Tukey–Kramer post hoc comparisons were used to look for differences between treatments within planned comparison groups. Leaf area was used to predict percent nitrogen in a simple linear regression model. All analyses were done using JMP version 3.2.6 (SAS Institute Inc. 1996).

Table 2. Sugar maple seedling means $\pm$ SE by treatment for leaf area, number of nodes, and total biomass.

Treatment	<i>n</i>	Leaf area (cm ²)	No. of nodes	Total biomass (g)
Control	17	107 $\pm$ 9	3.4 $\pm$ 0.3	0.98 $\pm$ 0.05
Average beech	16	77 $\pm$ 9	2.3 $\pm$ 0.4	0.85 $\pm$ 0.05
Average maple	17	102 $\pm$ 8	3.2 $\pm$ 0.3	0.91 $\pm$ 0.06
Average nutrient	17	114 $\pm$ 6	3.6 $\pm$ 0.3	1.04 $\pm$ 0.04
Dilute beech	17	103 $\pm$ 7	3.4 $\pm$ 0.3	1.02 $\pm$ 0.06
Dilute maple	17	104 $\pm$ 9	3.1 $\pm$ 0.3	0.94 $\pm$ 0.04
Dilute nutrient	17	113 $\pm$ 6	3.5 $\pm$ 0.3	0.97 $\pm$ 0.03

Results

Leaf area, development, and biomass

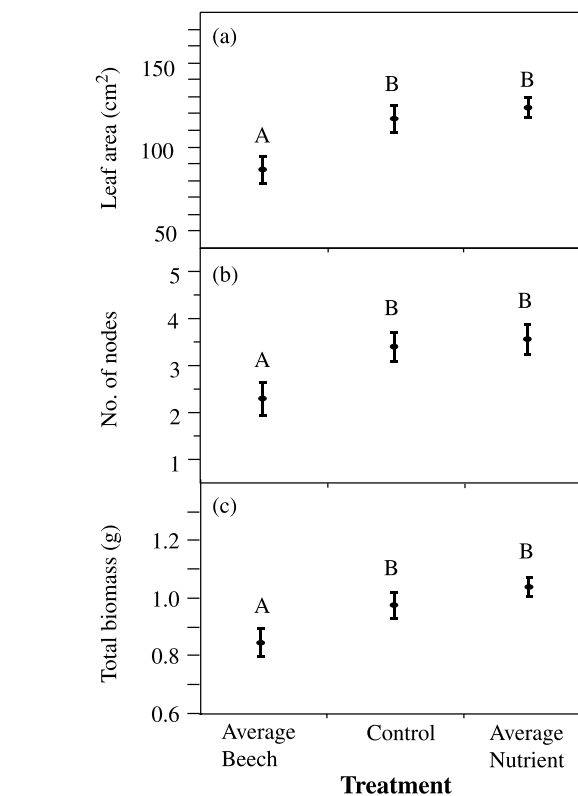
The average beech treatment had the lowest leaf area, the fewest nodes, and the lowest total biomass, while the average nutrient treatment had the highest leaf area, the most nodes, and the highest biomass, with the control being intermediate (Table 2). There were significant differences in leaf area ($p = 0.010$), number of nodes ($p = 0.021$), and biomass ($p = 0.005$) among average beech, average nutrient, and control treatments (Table 2). Results of the Tukey–Kramer test show that there is a significant difference between the average beech and control and average nutrient treatment responses for leaf area, number of nodes, and biomass (Fig. 1). Control and average nutrient treatments were not different from each other in any comparison.

For all other sets of planned comparisons (dilute beech, dilute nutrient, control; average maple, average nutrient, control; dilute maple, dilute nutrient, control), no significant differences among treatments were found for leaf area ($p = 0.22$), number of nodes ($p = 0.31$), and biomass ($p = 0.28$). Similar trends were found for the effect of leachate on leaf area for the dilute beech comparisons (dilute beech, dilute nutrient, control) as for the average beech treatment, but the differences were not significant (Table 2). For the maple treatments, there were no consistent trends, and no significant differences among treatments were found ($p = 0.21$) (Table 2).

Physiological traits

We tested physiological traits (carbon assimilation rate and stomatal conductance) of both the leaves at the first node and the leaves at subsequent nodes and found no significant difference among traits between nodes (t test, $p = 0.44$). We measured physiological traits of the leaves at the first node because all plants had leaves at that node but not all plants had leaves beyond the first node. Carbon assimilation rates for the sugar maple seedlings varied between treatments, with seedlings in the average beech treatment averaging $3.5 \pm 0.3 \mu\text{mol C}\cdot\text{m}^{-2}\cdot\text{s}^{-1}$ and the average nutrient plants averaging $2.4 \pm 0.3 \mu\text{mol C}\cdot\text{m}^{-2}\cdot\text{s}^{-1}$ ($p = 0.008$). Stomatal conductance also varied between these two treatments, with the average beech treatments having a significantly higher stomatal conductance than the plants in the average nutrient treatment ($p = 0.005$). However, there was no difference in water use efficiency ($p = 0.64$) between

these two treatments. Carbon flux for the whole plant (plant leaf area times assimilation rate) showed no differences between treatments; the plants in the average beech treatment had an average flux of $0.027 \pm 0.002 \mu\text{mol C}\cdot\text{s}^{-1}$ and the average nutrient plants an average flux of $0.026 \pm 0.002 \mu\text{mol C}\cdot\text{s}^{-1}$ ($p = 0.83$). The number of nodes had a negative linear relationship with carbon assimilation rate ($R^2 = 0.39$, $p < 0.001$), indicating that plants that had a higher number of nodes had a lower carbon assimilation rate.



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Plant tissue chemistry

The plant tissue chemistry differed among the treatments in the planned contrast of average beech, average nutrient, and control, with the beech treatment having significantly higher leaf nitrogen concentration ($p = 0.019$), higher stem nitrogen concentration ($p = 0.001$), lower stem calcium concentration ($p = 0.003$), lower stem magnesium concentration ($p = 0.008$), and lower stem potassium concentration ($p = 0.022$) than the other two treatments (Table 3). For all other sets of planned comparisons (dilute beech, dilute nutrient, control; average maple, average nutrient, control; dilute ma-

Table 3. Mean $\pm$ SD for sugar maple leaf, stem, and root nutrient concentration by treatment for nitrogen, calcium, magnesium, and potassium.

Treatment	<i>n</i>	N (%)	Ca (%)	Mg (%)	K (%)
Leaves					
Control	10	1.65 $\pm$ 0.06	0.69 $\pm$ 0.07	0.20 $\pm$ 0.02	1.16 $\pm$ 0.12
Average beech	10	1.86 $\pm$ 0.08	0.79 $\pm$ 0.04	0.20 $\pm$ 0.01	1.43 $\pm$ 0.12
Average maple	10	1.70 $\pm$ 0.04	0.74 $\pm$ 0.07	0.20 $\pm$ 0.02	1.46 $\pm$ 0.07
Average nutrient	10	1.63 $\pm$ 0.05	0.70 $\pm$ 0.10	0.17 $\pm$ 0.02	1.30 $\pm$ 0.16
Dilute beech	10	1.70 $\pm$ 0.06	0.68 $\pm$ 0.04	0.19 $\pm$ 0.02	1.36 $\pm$ 0.07
Dilute maple	10	1.77 $\pm$ 0.07	0.71 $\pm$ 0.06	0.19 $\pm$ 0.02	1.52 $\pm$ 0.10
Dilute nutrient	10	1.64 $\pm$ 0.06	0.62 $\pm$ 0.03	0.18 $\pm$ 0.01	1.29 $\pm$ 0.10
Stems					
Control	3	0.72 $\pm$ 0.03	0.64 $\pm$ 0.06	0.14 $\pm$ 0.02	1.13 $\pm$ 0.10
Average beech	3	1.14 $\pm$ 0.02	0.33 $\pm$ 0.06	0.07 $\pm$ 0.01	0.67 $\pm$ 0.09
Average maple	3	0.70 $\pm$ 0.02	0.66 $\pm$ 0.08	0.12 $\pm$ 0.01	1.11 $\pm$ 0.12
Average nutrient	3	0.74 $\pm$ 0.04	0.87 $\pm$ 0.05	0.16 $\pm$ 0.01	1.17 $\pm$ 0.11
Dilute beech	3	0.89 $\pm$ 0.16	0.66 $\pm$ 0.06	0.12 $\pm$ 0.01	1.08 $\pm$ 0.10
Dilute maple	3	0.93 $\pm$ 0.10	0.84 $\pm$ 0.16	0.16 $\pm$ 0.03	1.36 $\pm$ 0.10
Dilute nutrient	3	0.80 $\pm$ 0.01	0.56 $\pm$ 0.13	0.09 $\pm$ 0.02	0.86 $\pm$ 0.17
Roots					
Control	3	0.99 $\pm$ 0.01	0.06 $\pm$ 0.03	0.38 $\pm$ 0.09	1.29 $\pm$ 0.18
Average beech	3	1.22 $\pm$ 0.13	0.10 $\pm$ 0.02	0.29 $\pm$ 0.03	1.65 $\pm$ 0.16
Average maple	3	0.94 $\pm$ 0.13	0.14 $\pm$ 0.04	0.41 $\pm$ 0.07	1.80 $\pm$ 0.30
Average nutrient	3	0.93 $\pm$ 0.04	0.14 $\pm$ 0.05	0.43 $\pm$ 0.08	1.52 $\pm$ 0.15
Dilute beech	3	0.94 $\pm$ 0.06	0.11 $\pm$ 0.04	0.34 $\pm$ 0.04	1.53 $\pm$ 0.18
Dilute maple	3	1.01 $\pm$ 0.02	0.20 $\pm$ 0.06	0.43 $\pm$ 0.07	1.89 $\pm$ 0.09
Dilute nutrient	3	0.99 $\pm$ 0.05	0.11 $\pm$ 0.04	0.38 $\pm$ 0.06	1.34 $\pm$ 0.11

ple, dilute nutrient, control), no significant differences among treatments were found for leaf or stem chemistry (Table 3). The treatments showed no differences with regard to root chemistry (Table 3). On a mass basis, treatments did not show differences for stem nitrogen ($p = 0.47$), calcium ($p = 0.37$), magnesium ($p = 0.42$), or potassium ($p = 0.18$) or for leaf nitrogen ($p = 0.28$).

Discussion

Sugar maple seedling development was clearly affected by American beech leaf extract in this greenhouse experiment, as evidenced by differing seedling development rates as well as tissue chemistry among treatments. The seedlings treated with the higher-concentration beech leachate had smaller leaves, lower biomass, and fewer nodes than both the control and the average nutrient treatment (Fig. 1). Yet the control, average nutrient, and average sugar maple treatments did not show such trends, indicating the likely influence of secondary compounds (Table 2). The number of nodes in a plant is a direct measure of its developmental stage, and the seedlings in the average beech treatment had 30% fewer nodes than those in all of the other treatments. The carbon assimilation rate results also indirectly support the hypothesis that the seedlings in the average beech treatment were at a repressed developmental stage because seedlings at earlier stages of development have been reported to have higher rates of photosynthesis (Tichá et al. 1985; Sprugel 1990; Paembonan et al. 1992). The higher concentration of nitrogen in the leaves and stems of seedlings treated with average beech leachate (Table 3) may be a function of their younger

developmental stage as well. Plants are known to take up nitrogen and store it for future leaf expansion, causing a dilution effect when the nitrogen is spread out throughout the growing leaf (Luxmoore et al. 1981; Miller and Timmer 1997). Further evidence of this interpretation is illustrated by the negative relationship between leaf area and nitrogen concentrations. On a mass basis, there was no difference in the nitrogen in the leaves among treatments, so clearly, the plants in the average beech treatment group were taking up the nitrogen.

Other than the effects of average concentration beech leachate on sugar maple seedlings, treatments did not differ with regard to development or tissue chemistry concentrations. We did not observe a fertilizing effect of increased nutrients between treatments of dilute and average nutrient concentration, which may indicate that the concentrations of nutrients were not limiting plant growth at either concentration. The effects of the two maple treatments also did not differ from the effects of throughfall control or the nutrient controls, indicating that the compounds that suppressed sugar maple growth in the beech treatment were secondary compounds unique to the beech treatment and not generic DOC. The dilute concentration of beech leachate showed a slight effect on leaf area, but none of the effects differed statistically from the control treatment. This result suggested that a dose relationship existed in the response of the sugar maple seedlings to the beech treatment.

The possible mechanisms responsible for the interference of beech leachate with sugar maple development in this greenhouse study are unknown. Reduced rates of development in the seedlings treated with beech leachate may result

from interference with the hormones that control growth and development. Beech is known to produce a large number of secondary compounds (Coldwell and DeLong 1950; Smith 1976) including polyphenolics, which have been studied extensively in allelopathic relationships (Rice 1984; Einhellig 1995). Phenolic acids have multiple, generic actions on physiological functions (Einhellig 1995; Inderjit et al. 1999), although isolation and identification techniques that consider the full range of beech foliage natural products should be pursued. Further investigation should include the effects of natural products on both ion uptake and retention (Kobza and Einhellig 1987) and water relationships (Barkosky et al. 2000).

Traditionally, one of the criticisms of allelopathic research has been that it is difficult to distinguish between nutrient or pH effects and the effects of the allelochemicals themselves (Qasem and Hill 1989; Blum 1999). Our experiment was designed to compare treatments that had the same levels of inorganic solutes and pH but with a subset of treatments with varying levels of the leaf extracts in addition to the nutrients (average beech, dilute beech, average maple, dilute maple). Because we only saw an effect on seedling growth in the average beech treatment and not in the maple treatment, the logical conclusion is that compounds unique to the beech leaf extract had a negative impact on sugar maple seedling development and thus the chances of survival. An alternative explanation of the deleterious effect of beech leachate on sugar maple seedlings is that the increased microbial populations related to the leaf decomposition affected the sugar maples rather than the beech chemistry specifically. The maple treatment addresses this alternative to some degree, since that treatment would have increased microbial activity as well, and we did not see depressed development of sugar maple seedlings when they were treated with maple leaf leachate. We also attempted to limit microbial activity by keeping all leaf material and extracts at 4°C for the duration of the experiment.

Conditions in the greenhouse differ from those found in the field, particularly soil conditions. By using sterile potting soil, we were able to limit the likelihood that variation in soil microbial communities would affect the experimental results and confuse any allelochemical responses in the seedlings. In the field, differences in soil microbial communities may ameliorate or exacerbate any allelochemical effects and are often found to be a part of the allelochemical pathway of plant interference (Nilsson et al. 1993; Inderjit and Malik 1997). While we recognize that field soil would represent the ideal conditions for an allelopathy experiment, the logistics of this are formidable. While field soil would theoretically provide the ideal medium for planting the seedlings in the experiment, in practice, using field soil raises a number of logistic problems that made sterile potting soil a better choice. The most significant obstacle to using field soil in this experiment was that removing the soil from the field and putting it into pots while retaining the layers, density, and texture from the field proves to be impossible because of the number of rocks in the soil and the volume of soil needed to support a seedling. Mixing the soil would have precluded meaningful conclusions about chemical processes, since mixed soil does not behave the same way as layered soil, particularly with regard to nutrient cycling (Binkley and

Hart 1989). By mixing forest floor organic horizons with deeper mineral soil, a “priming effect” can be created, artificially stimulating soil microbial activity (Salonius 1978; Thorne and Hamburg 1985). Furthermore, the nutrient content, moisture, and organic matter vary tremendously with fine-scale microtopography at this site, making it difficult to ensure that soil conditions would be uniform across the experiment. Using preweighed sterile potting soil insured that each seedling was receiving soil of the same density and composition. This is an imperfect solution, since interactions in a natural environment may not occur in the same way as in a greenhouse study, but we consider this an important first step in elucidating a potential allelopathic effect.

Separating the effects of resource competition and allelopathy on in situ vegetation patterns is difficult to do because these two mechanisms may be acting in concert (Williamson 1990; Nilsson 1994; Inderjit and del Moral 1997). Recent studies have emphasized the importance of abiotic and biotic stress factors in the study of allelopathy and in particular the effects of nutrients, shade, plant diseases, and herbivores (Einhellig 1999). The strength of the effect of allelopathic chemicals on seedling growth is known to be affected by a variety of ecological factors, including soil type, temperature, pH, and microbial and mycorrhizal associations (Rice 1984; Nilsson et al. 1993; Nilsson 1994; Dakshini et al. 1999). We recognize that research that utilizes field soil conditions is required to demonstrate that allelopathy is a component of beech interference with sugar maple in the field. The present greenhouse experiment sets the stage for such work.

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